

CS 672: Advanced Topics in Deep Learning

Course Code	CS 672
Course Name	Advanced Topics in Deep Learning
L-T-P-C	3-0-2-4
Prerequisites	CS 671: Deep Learning and its Applications.
Distribution	Discipline Elective for final year BTech CSE, DSE and EE, Elective for other final year B.Tech disciplines, MS, M.Tech, and Ph.D. students.
Approval	recommended by the 38th BoA (Jan 2021)

Course Contents

- **GAN series:** Deep Convolutional GAN (DCGAN), Conditional GAN (cGAN), Wasserstein GAN (WGAN), Stacked GAN (StackGAN), Attention GAN, Picture to Picture GAN (Pix2Pix), Cyclic GAN (Cycle GAN), Discover Cross-Domain R-elations (DiscoGAN), Super Resolution GAN (SRGAN), Texture GAN, Self Attention GAN (SAGAN). [8 Lectures]
- **Transformer Networks:** Drawbacks of Recurrent Neural Networks, Self Attention, Transformers, Bidirectional Encoder Representation from Transformer (BERT), Generative pre-trained Transformer (GPT). [6 Lectures]
- **Deep Reinforcement Learning:** Basic of reinforcement learning, Markov decision process, Value and Q-value functions, Deep Q-learning, Deep Policy Gradient iteration (Reinforce Algo). [10 Lectures]
- **Graph-based Deep Learning:** Basics of Graph Convolutional Neural Network (GCN), Graph Embeddings, Spectral and Spatial GCNs, Graph Autoencoders. [5 Lectures]
- **Some latest miscellaneous deep learning paradigms and concepts:** [10 Lectures]
 - Capsule Network
 - Teacher-student network
 - Attention and Self-attention mechanism
 - Multi-task learning
 - Novel loss functions
 - Model compression/Network Pruning: redundant filter removal, filter ranking, and filter attention.
 - Explainable AI
- **Advance deep learning application** (optional/cover in above topics/related to projects): [3 Lectures]
 - CV related: Object detection, Tracking with Re-id, Flow networks,
 - NLP related: Summarization, text generation,
 - Misc: Domain Adaptation etc.

Reference Material

1. Most of the material will be covered from the recently published research papers at prestigious venues like NIPS, CVPR, ECCV, ICCV, ICLR, etc. [Lecture Material]
2. Aston Zhang et.al., **Dive into Deep Learning**, (Book website: <https://d2l.ai>) (Book PDF: <https://d2l.ai/d2l-en.ruti>)
3. Ian Goodfellow and Yoshua Bengio and Aaron Courville, **Deep Learning**, (Book website: [https://](https://www.deeplearningbook.org/)) [Reference Books]
4. [Few Reference Courses]
 - (a) [CS231n, DL Stanford course]
 - (b) [CS224n, NLP Stanford course]
 - (c) [DeepMind RL course by David Silver]